Exploratory data analysis of the mouse database

Background for the *Data* section of the paper. The notebook describes the whole database briefly (Part A), the experiment behind the study cohort in more detail (Part B, trail 3) and the study cohort itself in depth (Part C): the 186 mice and 3,049 daily weights that the results in `mouse_with_context.ipynb` are computed on. Part D collects the key numbers in one table.

The cohort is loaded with exactly the loader and configuration of the results notebook (`michael.load_dataset`, trail 3, roles RT/RChT/ChT), so every number here matches the results. The database is in German; all labels are translated in the setup cells below. Nothing is written to disk.

0. Setup

Translations of the German lookup tables

Role (`Rollen`), cause of death (`Todesursachen`) and killing method (`Toetungsarten`) are translated from the lookup tables; the trail names and aims from `Versuchsreihen`, the project titles from `Tierversuche`.

Plot style and helpers

Therapy groups keep one colour throughout the notebook (RT blue, RChT orange, ChT aqua); cell lines use three other hues so that they are never confused with the therapy groups.

Load the raw tables and the study cohort

```
Animals: 188 -> 186 after filtering -> 186 sampled
25 tables; cohort: 186 mice, 3049 rows
```

	item_id	timestamp	time	target	radio	chemo	intervention	trail	role
0	3	2013-08-05	13:00:00	32.5	0	0	0	3	1
1	3	2013-08-09	13:00:00	28.9	0	0	0	3	1
2	3	2013-08-13	13:30:00	29.0	0	0	0	3	1
3	3	2013-08-14	08:00:00	29.5	0	0	0	3	1
4	3	2013-08-19	08:39:00	29.4	0	0	0	3	1

Tidy English versions of the core tables

`mice` (one row per animal), `visits` (one row per visit) and one frame per treatment. These are only used for the description; the cohort `df` above comes straight from the loader.

	id	trail	role	role_abbr	sex	strain	batch	cohort	cell_line	birth	death	death_
0	1	3	5	E	male	PFP/RAG2	10	LNCaP RGB	LNCaP RGB	2013-10-12	2013-11-05	Experi ende pla
1	2	3	5	E	male	PFP/RAG2	10	LNCaP RGB	LNCaP RGB	2012-12-10	2013-11-05	Experi ende pla
2	3	3	1	RT	male	PFP/RAG2	10	LNCaP RGB	LNCaP RGB	2012-12-10	2013-11-05	Experi ende pla
3	4	1	2	RChT	male	PFP/RAG2	6	Vorversuch PC3	PC-3 RGB	2012-12-10	2014-12-18	Experi ende pla
4	5	1	2	RChT	male	PFP/RAG2	6	Vorversuch PC3	PC-3 RGB	2012-12-10	2014-12-15	Spontan d

Part A: the database (overview)

A1. Provenance and tables

The data come from the animal-experiment database of the radiotherapy research lab ("Mausprojekt"). The results use the pandas pickle `/data/mice/maus.pandas.pkl`, a conversion of the original MySQL dump (`maus_mysql.sql`, 2023). A newer dump (`mausprojekt-3.sql`, received 2025-10-02) exists but is **not** used here or in the results.

The pickle holds 25 tables. Two are the core of every analysis: `Versuchstiere` (one row per mouse) and `Visiten` (one row per visit with the body weight). Treatments, tumours and the experimental design hang off the mouse ID (`IdTier`):

```
Tierversuche (approved project, licence no.)
  └── Versuchsreihen (trail) ──────────────────────────
Versuchstiere.Versuchsreihe
Kohorten (trail × cell line × strain)
  └── VGruppe (injection batch) ──────────────────────
Versuchstiere.VGruppe
Versuchstiere (mouse, IdTier)
  ├── Visiten          weight, burden score sheet, tumour volume        1:n
  ├── Bestrahlungen    RT fractions            → RTKonzepte             1:n
  ├── Chemotherapie    ChT administrations     → ChTKonzepte, ChemoMed,
Applikation
  ├── Operationen, Narkosen, Prozeduren                                 1:n
  ├── Tumor            injection site, cell line, tumour weight →
Topographie, Tumorzelllinien
  └── lookups          Rollen (role), Todesursachen, Toetungsarten,
Tiertypen, Zuechter
```

	table	category	rows	columns	content
0	Visiten	core	22142	41	one row per visit: body weight, relative weight, burden score sheet (weight, general condition, behaviour, tumour), tumour volume
1	Versuchstiere	core	843	29	one row per mouse: trail, role (treatment group), sex, strain, birth and death date, cause of death, killing method, RT/ChT protocol, injection batch
2	Narkosen	treatment	2823	10	anaesthesia sessions (agent, gas flows, date)
3	Bestrahlungen	treatment	1958	18	one row per radiotherapy fraction: planned/delivered dose, field, date, time
4	Chemotherapie	treatment	1316	13	one row per chemotherapy administration: drug, protocol, dose, route, date
5	Operationen	treatment	47	9	surgeries (removal of the primary tumour), with tumour weight
6	Tumor	outcome	844	9	tumours per mouse: site, cell line, injection date, removal date, tumour weight
7	VGruppe	design	66	6	injection batches: cohort, cell line, date
8	Kohorten	design	27	6	cohorts: trail × cell line × mouse strain
9	Versuchsreihen	design	15	15	trails (experiment series): name, aims, start/end date, project
10	Tierversuche	design	12	11	approved animal-experiment projects with licence number
11	RTKonzepte	lookup	20	10	radiotherapy protocols: dose per fraction, total dose, fractions per day/week
12	Tumorzelllinien	lookup	19	6	tumour cell lines
13	ChTKonzepte	lookup	18	15	chemotherapy protocols: up to three drugs with dose, route and days
14	Topographie	lookup	15	4	anatomical sites (injection sites, organs at necropsy)
15	Rollen	lookup	11	3	roles = treatment groups (RT, RChT, ChT, controls, ...)
16	Tiertypen	lookup	9	7	mouse strains (and human as source of the tumour cells)
17	Todesursachen	lookup	8	3	causes of death
18	Applikation	lookup	6	3	administration routes
19	ChemoMed	lookup	6	7	chemotherapy drugs
20	Toetungsarten	lookup	5	3	killing methods
21	SeitenBez	lookup	4	2	body side

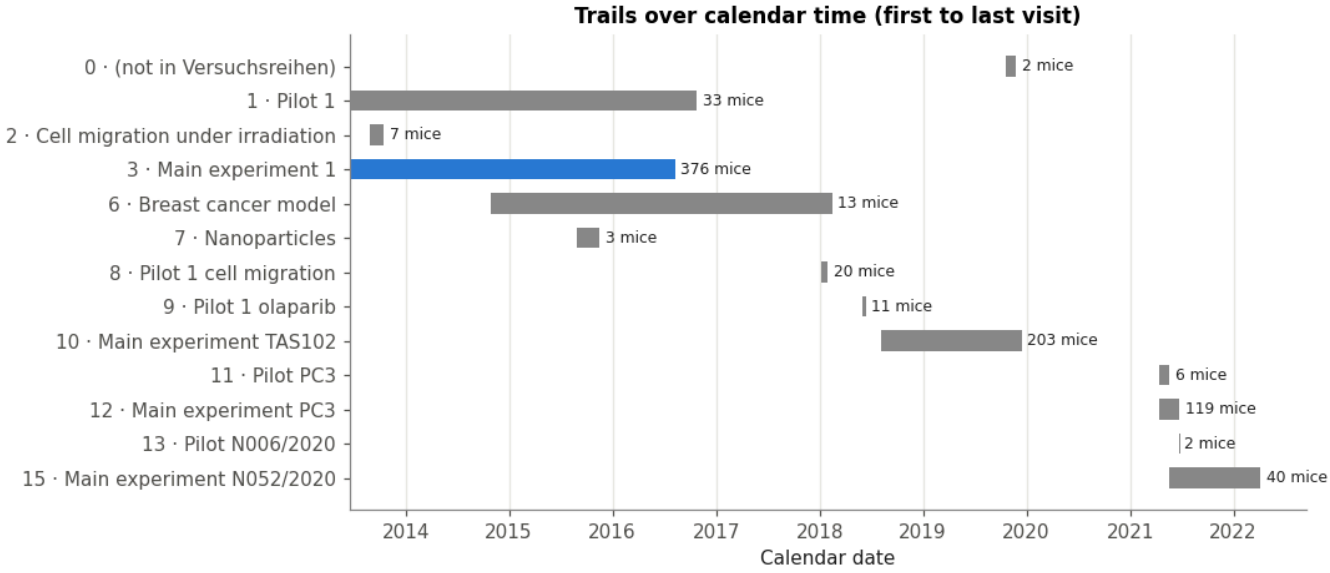
	table	category	rows	columns	content
22	Zuechter	lookup	4	6	breeders / suppliers of the mice
23	Prozeduren	notes	13	7	free-text clinical procedures (fluids, soft food, isolation, ...)
24	Tagescheck	admin	213	14	daily quality check of the irradiation device

A2. Projects and trails

A *trail* (`Versuchsreihe`) is one experiment series; several trails can belong to the same approved project (`Tierversuche` , with its licence number). Start/end are the dates stored for the trail; first/last visit are taken from the visits.

trail	name	aim	licence	project	start	end
0	(not in Versuchsreihen)	NaN	NaN	NaN	NaN	NaN
1	Pilot 1	dose finding; tolerability; tumour growth	G14/13	Influence of radiotherapy, chemotherapy, simultaneous radiochemotherapy and surgery on metastasis in xenograft mouse models	2013-06-17	None
2	Cell migration under irradiation	migration of in-vivo labelled cells into an irradiated area	G133/12	Selectin-dependent interaction of cell populations in a clinically relevant human tumour model under therapy	2013-07-24	None
3	Main experiment 1	effect of radiotherapy, chemotherapy, simultaneous radiochemotherapy and surgery on metastasis in xenograft mouse models; clonal selection of metastases	G14/13	Influence of radiotherapy, chemotherapy, simultaneous radiochemotherapy and surgery on metastasis in xenograft mouse models	2013-08-03	None
4	Pilot Loges 1	resistance and bone-marrow micrometastasis under anti-androgen therapy of breast cancer	G36/13	Resistance and bone-marrow micrometastasis under anti-androgen therapy of breast cancer	2013-11-27	None
5	Pilot 1 (methods)	learning methods; animal handling; tumour growth	G	Pilot with two animals: learning methods	None	None
6	Breast cancer model	tumour-specific radiosensitisation by targeted inhibition of DNA repair	NaN	NaN	2014-06-10	None
7	Nanoparticles	MRI of tumours; changes due to nanoparticles; changes due to local radiotherapy	G133/12	Selectin-dependent interaction of cell populations in a clinically relevant human tumour model under therapy	2015-08-27	None
8	Pilot 1 cell migration	migration of tumour cells; response to radiotherapy in the luciferase signal	G14/13	Influence of radiotherapy, chemotherapy, simultaneous radiochemotherapy and surgery on metastasis in	2018-01-08	None

	name	aim	licence	project	start	end trail
				xenograft mouse models		
9	Pilot 1 olaparib	effect of olaparib; test of cells and tissues; radiation response of normal tissue	G67/16	Radiosensitisation of PTEN- or CHD1-depleted prostate carcinomas by PARP1 inhibition	2018-05-31	None
10	Main experiment TAS102	efficacy of TAS102; tolerability; tumour growth	N013/2018	TAS102	2018-05-01	None
11	Pilot PC3	dose finding; tolerability; tumour growth	N127/2019	Influence of locally ablative radiotherapy, chemotherapy, radiochemotherapy and surgery on metastasis in two xenograft mouse prostate models	2021-04-13	2022-04-13
12	Main experiment PC3	effect of radiotherapy, chemotherapy, radiochemotherapy and surgery on metastasis; clonal selection of metastases	N127/2019	Influence of locally ablative radiotherapy, chemotherapy, radiochemotherapy and surgery on metastasis in two xenograft mouse prostate models	2021-04-16	None
13	Pilot N006/2020	dose finding; tolerability; method test	N006/2020	Immunomodulation by radiotherapy in glioblastoma	2021-06-22	None
14	Main experiment N006/2020	immunomodulation in glioblastoma	N006/2020	Immunomodulation by radiotherapy in glioblastoma	2021-06-23	2021-06-23
15	Main experiment N052/2020	response of metastases; response of brain tumours	N052/2020	Relevance of clonal heterogeneity for cerebral tropism and radiotherapy response of brain tumours	2021-09-17	2021-10-17



A3. Animals across trails

Counts of mice per trail by role (abbreviations as in `Rollen`: RT radiotherapy, RChT radiochemotherapy, ChT chemotherapy, K/E controls at the start/end, B observation without therapy, KT no local tumour or early termination, OP surgery, V pilot, – unassigned), sex, strain and cause of death.

role	meaning
0	Unassigned (role 0, not in lookup)
1	RT: radiotherapy
2	RChT: radiochemotherapy
3	ChT: chemotherapy
4	K: control animal at the start
5	E: control animal at the end
6	V: pilot experiment
7	B: observation without therapy
8	KT: no local tumour despite injection, or early termination
9	OP: surgical removal of the primary tumour
10	Abort: termination for general condition / spontaneous death
11	NNZ: not yet assigned

role trail	B	ChT	E	K	KT	OP	RChT	RT	V	–	Total
0	0	0	0	0	0	0	0	1	0	1	2
1	2	21	0	0	0	0	9	1	0	0	33
2	0	0	0	0	0	0	0	7	0	0	7
3	3	54	73	54	38	18	61	73	0	2	376
4	0	0	0	0	0	0	0	6	0	0	6
5	0	0	0	0	0	0	0	0	2	0	2
6	0	0	0	0	0	0	1	12	0	0	13
7	0	0	0	0	0	0	0	3	0	0	3
8	3	0	6	6	0	0	0	5	0	0	20
9	0	2	1	0	0	0	4	4	0	0	11
10	0	48	32	0	2	0	76	40	0	5	203
11	0	1	0	0	0	0	1	4	0	0	6
12	0	16	20	20	0	24	20	19	0	0	119
13	0	0	0	0	0	0	0	2	0	0	2
15	0	1	13	1	7	0	0	18	0	0	40
Total	8	143	145	81	47	42	172	195	2	8	843

sex trail	female	male	Total
0	1	1	2
1	8	25	33
2	0	7	7
3	139	237	376
4	6	0	6
5	2	0	2
6	12	1	13
7	3	0	3
8	0	20	20
9	0	11	11
10	112	91	203
11	0	6	6
12	0	119	119
13	0	2	2
15	39	1	40
Total	322	521	843

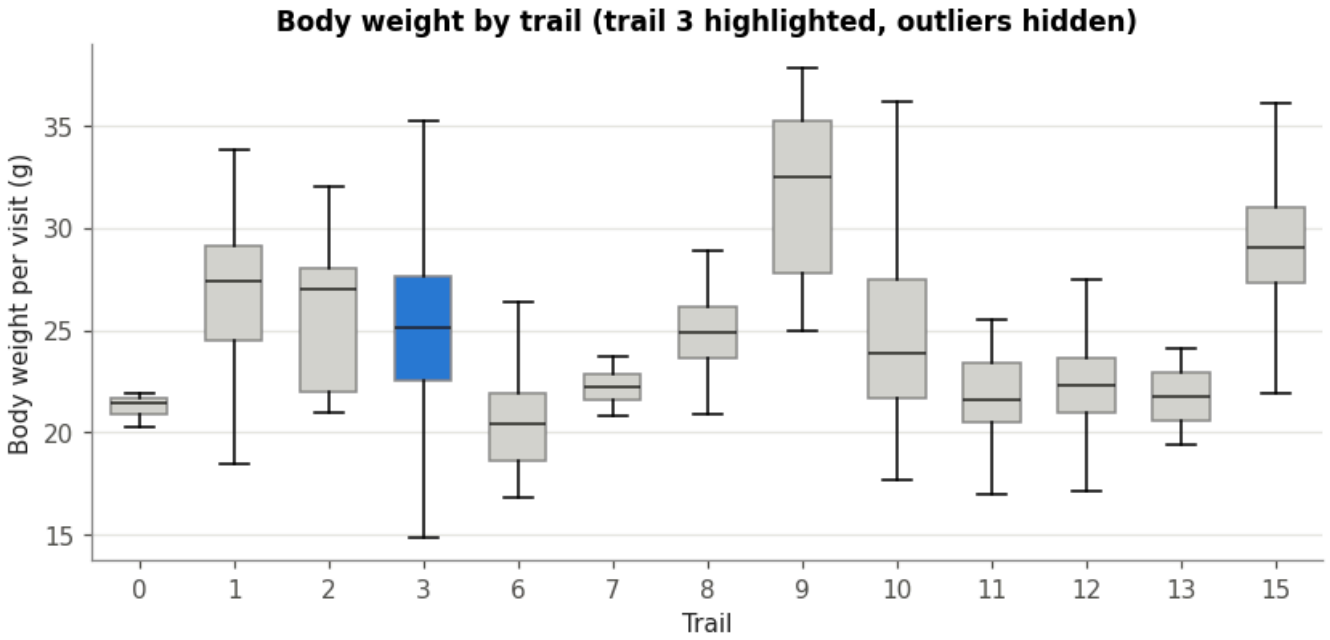
strain	Balb/c	C57Bl/6	FVB/N-Tg(MMTV-PyVT)	NMRI-Foxn1nu	NSG NOD.Cg-Prkdcscid	PFP/RAG2	SCID	SCID Select	Total
trail									
0	0	0	0	0	1	1	0	0	2
1	0	0	0	0	0	5	28	0	33
2	0	0	0	0	0	0	4	3	7
3	0	0	0	0	0	95	281	0	376
4	6	0	0	0	0	0	0	0	6
5	0	0	0	0	0	2	0	0	2
6	0	0	12	0	0	0	1	0	13
7	0	0	0	0	0	3	0	0	3
8	0	0	0	0	0	0	20	0	20
9	0	0	0	0	11	0	0	0	11
10	0	0	0	0	0	203	0	0	203
11	0	0	0	0	0	0	6	0	6
12	0	0	0	0	0	0	119	0	119
13	0	2	0	0	0	0	0	0	2
15	0	0	0	39	0	0	1	0	40
Total	6	2	12	39	12	309	460	3	843

death_cause	(no death record)	Death directly after surgery	Experiment ended as planned	Killed: poor general condition	Killed: tumour ulceration	Poor general condition / suspected lymphoma	Spontaneous death	Unknown	Unkno dispo o ani
trail									
0	0	0	1	0	0	0	0	1	
1	2	0	3	12	1	4	8	3	
2	7	0	0	0	0	0	0	0	
3	0	2	296	40	1	4	11	17	
4	6	0	0	0	0	0	0	0	
5	0	0	1	0	1	0	0	0	
6	0	0	12	0	0	0	0	1	
7	0	0	3	0	0	0	0	0	
8	0	0	20	0	0	0	0	0	
9	0	0	11	0	0	0	0	0	
10	8	0	169	0	25	0	0	1	
11	0	0	6	0	0	0	0	0	
12	0	3	105	5	0	0	6	0	
13	2	0	0	0	0	0	0	0	
15	38	0	2	0	0	0	0	0	
Total	63	5	629	57	28	8	25	23	

A4. Measurements and treatments across trails

Visits per trail and per mouse and the body weight per visit. Weights of 0 g and above 100 g are entry errors (see A5) and are left out of the boxplot.

trail	visits	mice_with_visits	visits_per_mouse	weight_g
0	35	1	35 [35–35]	21.4 [20.9–21.7]
1	1217	33	25 [7–70]	27.4 [24.5–29.1]
2	7	7	1 [1–1]	27.0 [22.0–28.0]
3	7047	376	15 [10–19]	25.1 [22.5–27.6]
6	13	13	1 [1–1]	20.4 [18.6–21.9]
7	39	3	13 [13–13]	22.2 [21.6–22.9]
8	428	20	23 [19–24]	24.9 [23.6–26.1]
9	11	11	1 [1–1]	32.5 [27.8–35.2]
10	7159	202	33 [25–43]	23.9 [21.7–27.5]
11	192	6	30 [29–34]	21.6 [20.5–23.4]
12	4705	119	41 [37–42]	22.3 [21.0–23.6]
13	2	2	1 [1–1]	21.8 [20.6–22.9]
15	1286	40	16 [13–38]	29.0 [27.3–31.0]



Trail 3 is by far the largest trail: about 45% of all mice and a third of all visits. Trails 10 (TAS102) and 12 (PC3) are the next largest; most other trails are small pilots.

trail	RT fractions	mice with RT	ChT administrations	mice with ChT	surgeries	anaesthesias	RT protocols	ChT drugs
0	3	1	0	0	0	3	d1-5 2.0 Gy	
1	36	8	84	28	0	97	d1-5 10,0 Gy	Cisplatin, Docetaxel, Etoposid
2	7	7	0	0	0	7	d1 10,0 Gy, d1 15,0 Gy	
3	666	134	192	116	19	867	d1-5 10,0 Gy	Cisplatin, Docetaxel
4	9	6	0	0	0	9	d1 6 Gy - Femora, d1 6 Gy - Mamma	
6	12	12	0	0	0	12	Mamma01 d1 3,0 Gy, Mamma02 d1 10,0 Gy	
7	15	3	0	0	0	15	d1-5 10,0 Gy	
8	15	5	0	0	0	15	d1 10,0 Gy, d1-2 10.0 Gy, d1-3 10.0 Gy, d1-4 10.0 Gy, d1-5 10,0 Gy	
9	8	8	6	6	0	8	d1 2.0 Gy	Olaparib
10	816	116	958	123	0	1359	d1 2.0 Gy, d1 3.0 Gy, d1-10 3,0 Gy, d1-5 2.0 Gy	Capecitabine, TAS102
11	13	5	3	2	0	16	d1-5 10,0 Gy, d1-5 10,0 Gy 225 kV	Docetaxel
12	193	39	71	36	28	246	d1-5 10,0 Gy, d1-5 10,0 Gy 225 kV	Docetaxel
13	2	2	0	0	0	2	d1 10 Gy225 kV	
15	163	18	2	1	0	167	d1-10 2.0 Gy, d1-11 12x2,0 Gy, d1-6 6x2,0 Gy	Docetaxel

A5. Data-quality issues in the database

Counted over the whole database. Most of them are removed by the cohort preprocessing (Part C); the column *in trail 3, RT/RChT/ChT* shows how many concern the 188 candidate mice of the study cohort.

	issue	unit	count	in trail 3, RT/RChT/ChT
0	Mice in trail 0, which is not in Versuchsreihen	mice	2	0
1	Mice without a death date	mice	63	0
2	Mice with an injection date before birth	mice	2	0
3	Visits with an invalid date (e.g. 0000-00-00)	visits	5	0
4	Visits of a mouse not in Versuchstiere	visits	1	0
5	Visits before the birth of the mouse	visits	29	0
6	Visits before the start of the trail	visits	139	36
7	Visits with weight 0 g	visits	10	1
8	Visits with weight > 100 g	visits	11	1
9	Visits with tumour volume > 100 cm^3	visits	6	1
10	Visits where the total score is not the sum of the sub-scores	visits	24	0
11	Mouse-days with more than one visit	mouse-days	304	10
12	Trails without a start date	trails	1	0
13	Trails without mice	trails	1	0

Part B: trail 3, the experiment behind the cohort

B1. Design

Trail 3 ("Hauptversuch 1", main experiment 1) belongs to project G14/13. Human tumour cells were injected subcutaneously into immunodeficient mice (xenografts). The mice were then allocated to radiotherapy, chemotherapy, radiochemotherapy, surgery or control groups, and followed with regular visits until they were killed. The cell lines carry an RGB marking (red/green/blue fluorescent labels), which serves the second aim of the trail, the clonal selection of metastases.

trail	**3**
name	Main experiment 1
aim	effect of radiotherapy, chemotherapy, simultaneous radiochemotherapy and surgery on metastasis in xenograft mouse models; clonal selection of metastases
licence	G14/13
project	Influence of radiotherapy, chemotherapy, simultaneous radiochemotherapy and surgery on metastasis in xenograft mouse models
start	2013-08-03
first_visit	2013-06-17 00:00:00
last_visit	2016-08-08 00:00:00
mice	376
visits	7047

	IdKohorte	KohorteName	cell_line	tumour_type	host_strain	mice	Bemerkung
6	7	OH1 RGB	OH-1 RGB	small-cell lung cancer (SCLC)	SCID	175	Hauptversuch OH1 RGB
7	8	H69 RGB	H69 RGB	small-cell lung cancer (SCLC)	SCID	103	Hauptversuch H69 RGB
8	9	LNCaP RGB	LNCaP RGB	prostate cancer (androgen-sensitive)	PFP/RAG2	95	Hauptversuch LNCaP RGB
9	10	PC3 RGB	PC-3 RGB	prostate cancer (androgen-independent)	PFP/RAG2	0	Hauptversuch PC3 RGB
10	11	MRT-Kontrolltiere	NaN	NaN	NaN	2	Tiere ohne Tumorzellen

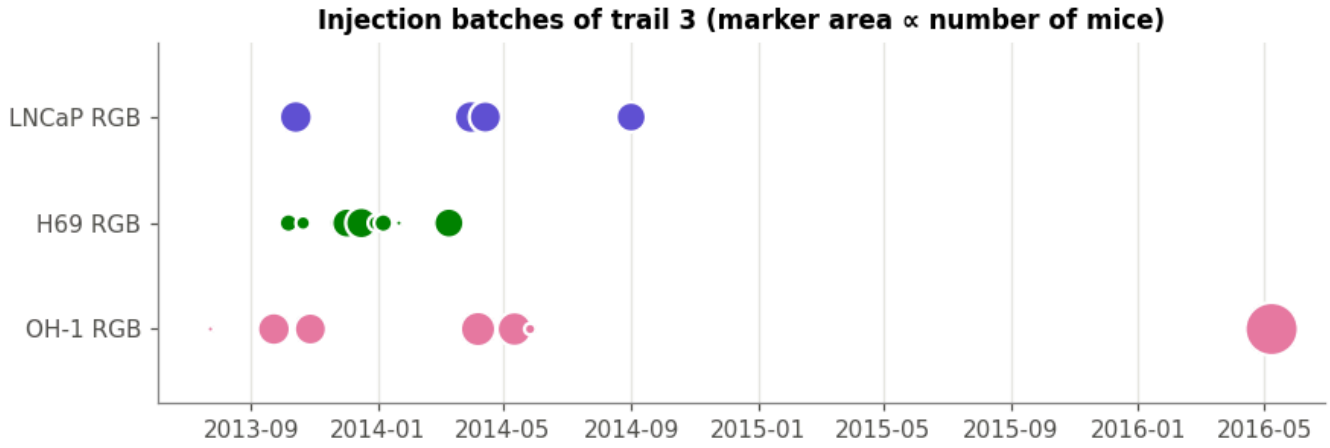
Mice of trail 3 outside these cohorts belong to injection batches of other cohorts (see the table below). Injection site and number of tumour records per mouse:

	tumour records
site	
s.c., right flank (injection)	323
s.c., right side, end of thorax (injection)	70

tumour records per mouse	mice
0	4
1	351
2	21

batch	cohort	cell_line	mice	roles	date	remark
57	Kooperation Zellmigration	OH-1 RGB	1	KT 1	2013-07-24	Vorzeitiger Therapieabbruch
8	OH1 RGB	OH-1 RGB	25	ChT 5, RChT 5, RT 5, E 5, K 4, B 1	2013-09-23	Hauptversuch OH1 01
9	H69 RGB	H69 RGB	8	RT 2, E 2, RChT 2, K 1, ChT 1	2013-10-07	Hauptversuch H69 01
10	LNCaP RGB	LNCaP RGB	25	E 5, RT 5, K 5, ChT 5, RChT 5	2013-10-14	Hauptversuch LNCaP 01
11	H69 RGB	H69 RGB	5	RChT 2, RT 1, E 1, ChT 1	2013-10-21	Hauptversuch H69 02
12	OH1 RGB	OH-1 RGB	24	E 6, RChT 5, ChT 5, RT 5, K 3	2013-10-28	Hauptversuch OH1 02
13	H69 RGB	H69 RGB	10	ChT 2, RT 2, E 2, RChT 2, K 2	2013-11-25	Hauptversuch H69 03
14	H69 RGB	H69 RGB	21	E 6, ChT 4, RT 4, RChT 4, K 3	2013-12-02	Hauptversuch H69 04
15	H69 RGB	H69 RGB	23	E 5, ChT 5, RT 5, RChT 5, K 3	2013-12-16	Hauptversuch H69 05
16	H69 RGB	H69 RGB	6	E 6	2013-12-30	Hauptversuch H69 06 Kontrolltiere
17	H69 RGB	H69 RGB	8	RChT 2, RT 2, ChT 2, E 1, K 1	2014-01-06	Hauptversuch H69 07
18	H69 RGB	H69 RGB	1	E 1	2014-01-21	Hauptversuch H69 08 Kontrolltier
19	H69 RGB	H69 RGB	21	KT 21	2014-03-10	Hauptversuch H69 kein lokaler TU trotz Injektion oder vorzeitiger Abbruch
20	LNCaP RGB	LNCaP RGB	25	K 5, RT 5, RChT 5, E 5, ChT 5	2014-03-31	Hauptversuch LNCaP 02
21	MRT-Kontrolltiere	None	2	B 2	2014-03-31	MRT Kontrolltiere
23	OH1 RGB	OH-1 RGB	29	RChT 8, E 6, K 5,	2014-04-07	Hauptversuch OH1 03

batch	cohort	cell_line	mice	roles	date	remark
				ChT 5, RT 5		
24	LNCaP RGB	LNCaP RGB	24	RT 5, RChT 5, ChT 5, E 5, K 4	2014-04-14	Hauptversuch LNCaP 03
25	OH1 RGB	OH-1 RGB	28	RChT 6, RT 5, E 5, ChT 5, K 5, - 2	2014-05-12	Hauptversuch OH1 04
26	OH1 RGB	OH-1 RGB	3	KT 3	2014-05-27	Hauptversuch OH1 kein TU trotz lokaler Injektion
27	LNCaP RGB	LNCaP RGB	21	RChT 5, RT 4, K 4, E 4, ChT 4	2014-09-01	Hauptversuch LNCaP 04
30	OH1 RGB	OH-1 RGB	66	OP 18, RT 18, KT 13, K 9, E 8	2016-05-08	Hauptversuch OH1 mit Operation



Three human cell lines were used: the two small-cell lung cancer lines OH-1 and H69 in SCID mice and the prostate cancer line LNCaP in (male) PFP/RAG2 mice. The PC-3 cohort of the trail has no mice. Most batches were injected between September 2013 and September 2014. One late OH-1 batch followed in May 2016 ("main experiment OH1 with surgery").

B2. Group allocation

The role of a mouse is its treatment group. Only the three treated groups RT, RChT and ChT are in the study cohort. The others are left out because they received no radio- or chemotherapy (controls K and E, observation B), had surgery

instead (OP), did not grow a local tumour or were terminated early (KT), or were never assigned (role 0).

cell_line role	(none)	H69 RGB	LNCaP RGB	OH-1 RGB	Total
B	2	0	0	1	3
ChT	0	15	19	20	54
E	0	24	19	30	73
K	0	10	18	26	54
KT	0	21	0	17	38
OP	0	0	0	18	18
RChT	0	17	20	24	61
RT	0	16	19	38	73
–	0	0	0	2	2
Total	2	103	95	176	376

sex role	female	male	Total
B	0	3	3
ChT	18	36	54
E	28	45	73
K	18	36	54
KT	17	21	38
OP	8	10	18
RChT	19	42	61
RT	29	44	73
–	2	0	2
Total	139	237	376

strain role	PFP/RAG2	SCID	Total
B	0	3	3
ChT	19	35	54
E	19	54	73
K	18	36	54
KT	0	38	38
OP	0	18	18
RChT	20	41	61
RT	19	54	73
–	0	2	2
Total	95	281	376

strain cell_line	PFP/RAG2	SCID	Total
(none)	0	2	2
H69 RGB	0	103	103
LNCaP RGB	95	0	95
OH-1 RGB	0	176	176
Total	95	281	376

sex cell_line	female	male	Total
(none)	0	2	2
H69 RGB	49	54	103
LNCaP RGB	0	95	95
OH-1 RGB	90	86	176
Total	139	237	376

B3. Treatment protocols

Radiotherapy

Planned protocol per mouse (`Versuchstiere.RTKonzept`) and the fractions actually recorded in `Bestrahlungen`.

IdRTKonzept	RTKonzept	dose per fraction (Gy)	total dose (Gy)	fractions	fractions/day	fractions/week	beam	Bemerkung
1	d1-5 10,0 Gy	10	50	5	1	5	200 kV	Standard 5 x 10 Gy

role_abbr	mice	with_RT	fractions	total_dose_Gy	days_first_to_last_fraction
B	3	0	{}	{}	{}
ChT	54	0	{}	{}	{}
E	73	0	{}	{}	{}
K	54	0	{}	{}	{}
KT	38	0	{}	{}	{}
OP	18	0	{}	{}	{}
RChT	61	61	{5.0: 61}	{50.0: 61}	{4.0: 61}
RT	73	73	{1.0: 1, 5.0: 72}	{10.0: 1, 50.0: 72}	{0.0: 1, 4.0: 72}
–	2	0	{}	{}	{}

Delivered dose per fraction (Gy): {10: 666}

Chemotherapy

Drug and protocol per cell line, number of administrations per mouse and the timing relative to the first radiotherapy fraction for the RChT group.

IdChTKonzept	protocol	dose (mg/kg)	days	Bemerkung
2	Doctaxel 20 mg/kg d1-8-15	20	3	None
5	Cisplatin 6mg/kg d1	6	1	

cell_line	drug	role	ChT (administrations)	RChT (administrations)	RT (administrations)	Total (administrations)
H69 RGB	Cisplatin		15	17	0	32
LNCaP RGB	Docetaxel		57	58	1	116
OH-1 RGB	Cisplatin		20	24	0	44
Total			92	99	1	192

role_abbr	drug	mice	administrations	route
ChT	**Cisplatin**	35	{1.0: 35}	i.p.
	Docetaxel	19	{3.0: 19}	i.p.
RChT	**Cisplatin**	41	{1.0: 41}	i.p.
	Docetaxel	20	{2.0: 2, 3.0: 18}	i.p.
RT	**Docetaxel**	1	{1.0: 1}	i.p.

RChT: days from the first RT fraction to the first ChT administration: {0: 61}

Protocols in short.

- **RT:** 5 × 10 Gy (50 Gy) to the tumour on five consecutive days (200 kV).
- **ChT:** depends on the tumour type. The SCLC lines (OH-1, H69) got cisplatin 6 mg/kg i.p. once; LNCaP got docetaxel 20 mg/kg i.p. on days 1, 8 and 15.
- **RChT:** both. Chemotherapy starts on the day of the first RT fraction (simultaneous).

Deviations from the planned protocol

Mice whose recorded treatment does not match their role.

	id	role_abbr	cell_line	fractions	drug	administrations	death_cause	in_study_cohort	
12	13	RT	LNCaP RGB	5.0	Docetaxel	1.0	Experiment ended as planned	True	ve
159	160	RT	H69 RGB	1.0	NaN	NaN	Killed: poor general condition	True	

Translation of the notes: mouse 13 "received ChT once by mistake on 21 Oct 2013"; mouse 160 "increasing paralysis of the legs" (only 1 of 5 fractions). Both stay in the study cohort, as RT.

Anaesthesia

Recorded anaesthesia sessions per mouse, by role.

Narkotikum role	CO2O2	Isofluran	Rompun/Ketanest i.p.	Total
ChT	33	54	0	87
E	0	0	1	1
OP	0	0	18	18
RChT	32	367	0	399
RT	1	361	0	362
Total	66	782	19	867

	sessions per mouse
role_abbr	
ChT	1 [1–3]
E	1 [1–1]
OP	1 [1–1]
RChT	6 [6–7]
RT	5 [5–5]

B4. Ages

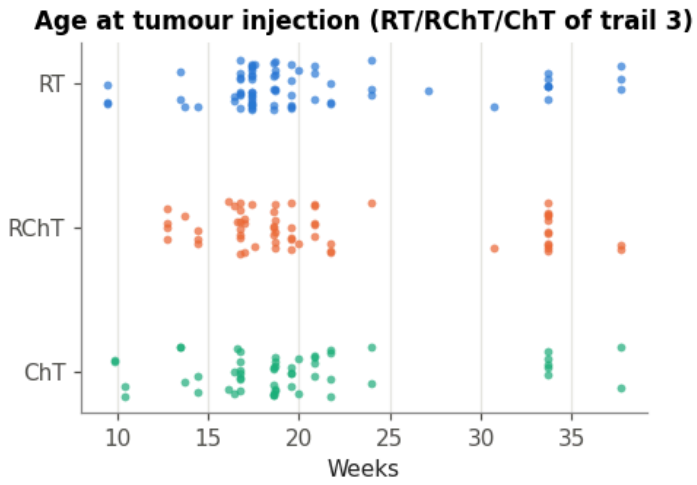
Age at tumour injection, at the first visit and at death, in weeks. The age at injection is the usual way to describe the animals in a methods section.

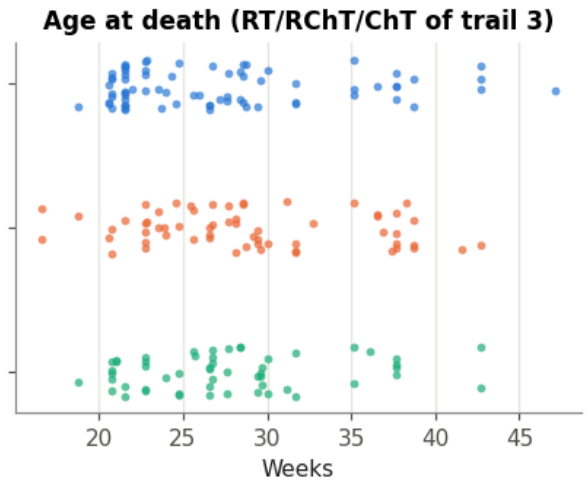
	mice	age_at_injection_wk	age_at_first_visit_wk	age_at_death_wk
role_abbr				
B	3	16.7 [16.7–16.7]	27.9 [22.2–27.9]	27.9 [23.2–27.9]
ChT	54	18.7 [16.7–20.9]	18.6 [16.6–20.7]	26.7 [22.7–29.9]
E	73	18.6 [16.7–20.9]	18.6 [15.6–20.7]	25.1 [22.6–29.6]
K	54	18.6 [17.0–20.9]	18.4 [15.6–20.7]	24.1 [20.7–29.0]
KT	38	17.4 [16.7–18.6]	16.6 [15.6–18.6]	28.3 [24.4–55.3]
OP	18	17.4 [17.4–17.4]	15.6 [15.6–15.6]	21.6 [20.4–21.6]
RChT	61	18.7 [16.7–21.7]	18.6 [16.6–20.7]	28.1 [23.9–32.7]
RT	73	18.6 [17.4–20.9]	18.4 [15.6–20.6]	25.6 [21.6–29.6]
–	2	37.7 [37.7–37.7]	37.6 [37.6–37.6]	40.1 [39.8–40.3]

cell_line	mice	age_at_injection_wk	age_at_death_wk
H69 RGB	103	19.6 [18.6–19.6]	26.6 [24.7–28.9]
LNCaP RGB	95	16.6 [14.0–21.7]	28.4 [24.6–31.4]
OH-1 RGB	176	17.4 [17.4–33.7]	22.7 [20.7–36.2]

Implausible ages at injection (< 3 weeks):

	id	role_abbr	birth	injection	first_visit	death
0	1	E	2013-10-12	2013-06-18	2013-06-17	2013-11-05





B5. Outcomes: causes of death and time on study

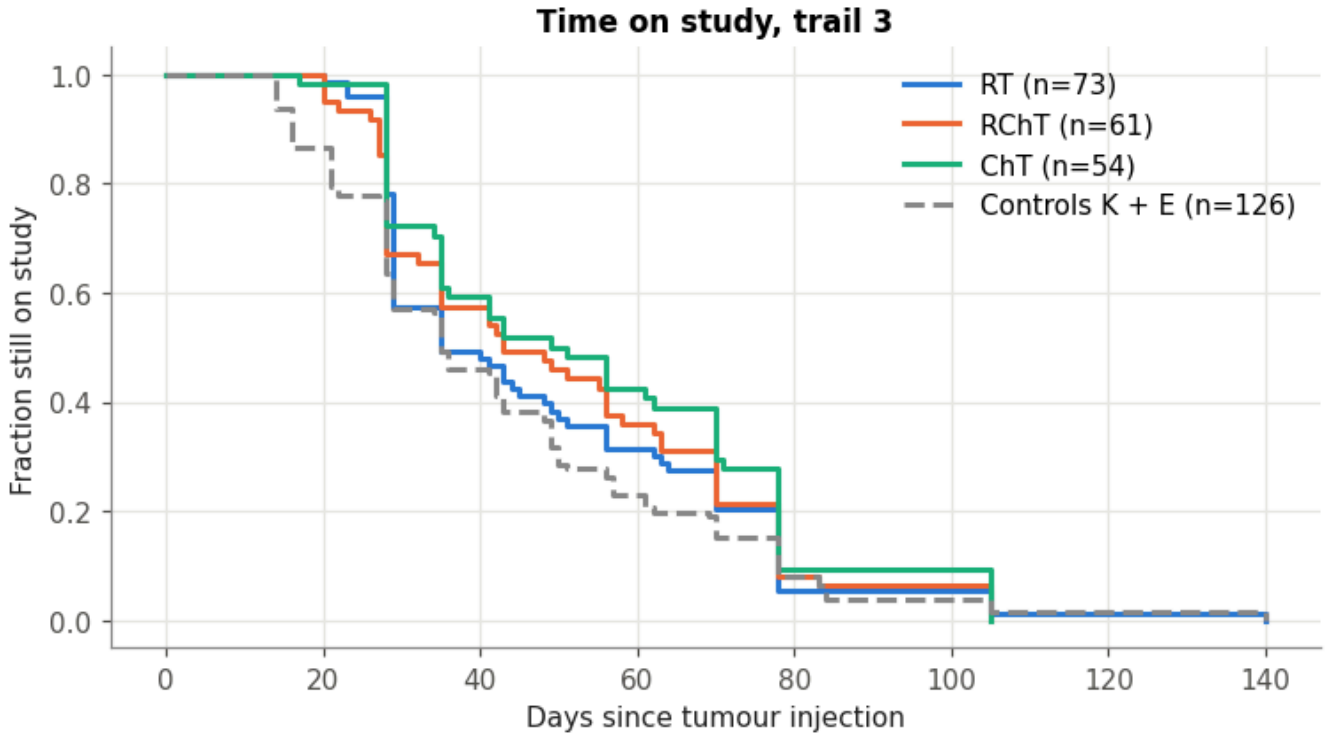
In trail 3 almost every mouse was killed at a planned endpoint ("experiment ended as planned"), so the end of a weight series is mostly set by the protocol, not by the disease. Few mice died spontaneously or were killed early because of their condition.

role death_cause	B	ChT	E	K	KT	OP	RChT	RT	–	Total
Death directly after surgery	0	0	0	0	0	2	0	0	0	2
Experiment ended as planned	0	52	66	54	10	12	42	60	0	296
Killed: poor general condition	2	0	7	0	16	4	6	5	0	40
Killed: tumour ulceration	0	1	0	0	0	0	0	0	0	1
Poor general condition / suspected lymphoma	0	0	0	0	4	0	0	0	0	4
Spontaneous death	1	1	0	0	2	0	4	3	0	11
Unknown	0	0	0	0	6	0	4	5	2	17
Unknown: disposed of by animal care	0	0	0	0	0	0	5	0	0	5
Total	3	54	73	54	38	18	61	73	2	376

role kill_method	B	ChT	E	K	KT	OP	RChT	RT	–	Total
CO2	0	0	4	1	0	0	0	0	0	5
Cervical dislocation	2	0	0	0	4	0	0	1	0	7
Final blood sampling	0	50	66	47	26	16	45	62	0	312
Final ketamine anaesthesia	0	3	3	5	0	0	3	2	0	16
Spontaneous death (not killed)	1	1	0	1	8	2	13	8	2	36
Total	3	54	73	54	38	18	61	73	2	376

```
'296 (79%)'
```

Time on study from tumour injection to death. Because nearly all deaths are planned, the Kaplan–Meier curve shows how long the groups were followed, **not** survival in a clinical sense.



role_abbr	days from injection to death
B	13 [13–13]
ChT	50 [28–78]
E	46 [29–70]
K	22 [16–42]
KT	74 [64–268]
OP	29 [21–29]
RChT	43 [28–70]
RT	35 [29–70]
–	16 [15–18]

The steps at about 28, 35, 42, 56, 70, 78 and 105 days are the planned end dates of the injection batches: the curves show the follow-up design more than the course of the disease.

Tumour weight at necropsy (sum over the primary tumour records of a mouse). The tumour is part of the measured body weight.

role_abbr	tumour weight
B	0.00 [0.00–0.00] g
ChT	0.18 [0.01–0.49] g
E	0.38 [0.05–0.99] g
K	0.13 [0.01–0.27] g
KT	0.05 [0.00–0.49] g
OP	0.09 [0.04–0.17] g
RChT	0.02 [0.00–0.08] g
RT	0.03 [0.01–0.11] g
–	0.00 [0.00–0.00] g

cell_line	tumour weight
H69 RGB	0.15 [0.00–0.66] g
LNCaP RGB	0.01 [0.00–0.09] g
OH-1 RGB	0.12 [0.04–0.40] g



Tumours of irradiated mice (RT, RChT) are much smaller at necropsy than those of the untreated controls at the end (E). Most tumours weigh well below 1 g, so at most a few percent of the body weight, but single tumours reach about 2 g, close to 10% of a 25 g mouse.

B6. Clinical burden score sheet

At every visit the animal is scored on a burden score sheet (German *Score-Sheet*), with four parts:

Column	Part	Items
`KGPunkte`	body weight	loss relative to the reference weight (`KGVerlauf`, %)
`AZPunkte`	general condition	fur, body orifices, eyes, posture, muscle tone, hydration, cramps, paralysis, temperature, breathing
`VHPunkte`	behaviour	sleep, reaction to blowing and touch, curiosity, social contact, motor function, isolation, pain, apathy, coordination, self-mutilation
`TUPunkte`	tumour	tumour volume and ulceration
`Score`	total	sum of the four parts

The weight and tumour points follow fixed thresholds, which can be read off the data:

weight points	relative weight % (count)	relative weight % (min)	relative weight % (median)	relative weight % (max)
0.0	6297	0.000000	101.937984	949.572650
10.0	312	80.097087	86.200463	89.958159
20.0	74	56.310680	76.995977	79.912664

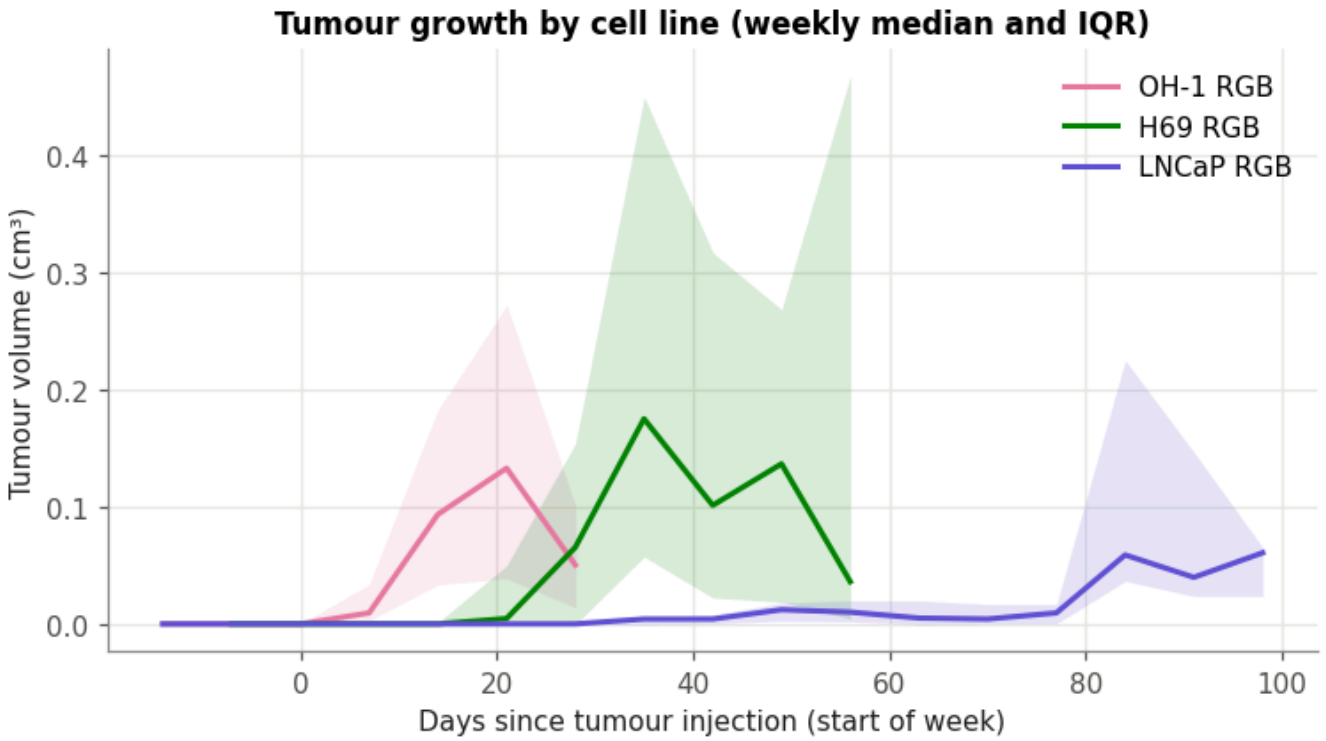
tumour points	visits	min_volume	max_volume	ulcerated
0	4025	0.0000	0.0000	0
5	2996	0.0001	1.0000	0
10	23	1.0800	2.7000	0
25	1	0.0838	0.0838	1

So the weight points are 0 at ≥ 90% of the reference weight, 10 at 80–90% and 20 below 80% (more than 20% weight loss). The tumour points are 5 up to a volume of 1 and 10 above, so the volume is recorded in cm^3 (ml). A total of 20 points is the level at which such score sheets usually require ending the experiment; check the threshold in the approved protocol before citing it.

role_abbr	mice	max_score	reached_score_20	reached_weight_points_10	reached_weight_points_20
B	3	25 [18–25]	2 (67%)	1 (33%)	0 (0%)
ChT	54	5 [5–15]	4 (7%)	20 (37%)	2 (4%)
E	73	5 [5–5]	7 (10%)	4 (5%)	1 (1%)
K	54	5 [5–5]	2 (4%)	5 (9%)	0 (0%)
KT	38	10 [1–10]	7 (18%)	15 (39%)	2 (5%)
OP	18	5 [5–10]	2 (11%)	5 (28%)	2 (11%)
RChT	61	15 [15–20]	18 (30%)	47 (77%)	18 (30%)
RT	73	15 [5–20]	21 (29%)	37 (51%)	17 (23%)
–	2	5 [2–8]	0 (0%)	1 (50%)	0 (0%)

The weight part dominates the burden of the treated mice. In the RChT group three in four mice lost at least 10% of their reference weight at some visit, and almost a third reached 20 points in total.

Tumour volume over time since injection, median and interquartile range per cell line (RT/RChT/ChT mice). Volumes are averaged per mouse and week first; weeks with fewer than 8 mice are not drawn.



Part C: the study cohort used for the results

C1. Cohort derivation

The flow below replays `michael.load_dataset` step by step with the configuration of the results notebook, so the counts end exactly at the data the models see. First the animals, then the rows.

	step	mice
0	All animals in the database	843
1	in a trail listed in Versuchsreihen	841
2	in trail 3 (Main experiment 1)	376
3	with role RT, RChT or ChT	188
4	with a death date	188
5	passing the series filters below (study cohort)	186

Excluded at the last step:

id	treated (RT or ChT recorded)	at least 5 weights	all weights > 0 g	all weights ≤ 100 g	n	min_weight	max_weight	role
82	True	True	False	True	16	0.0	20.1	RChT
313	True	True	True	False	10	21.5	222.2	ChT

step	visit	radio	chemo	surgery	birth	death	with weight
Records of the cohort mice (visits, treatments, birth, death)	3094	661	190	0	186	186	3094
after dropping records before birth / before the trail start (2013-08-03) or without a valid date	3058	661	190	0	70	186	3058
each treatment takes over the weight of the nearest visit, which is removed	2268	661	190	0	70	186	3119

```
Dropped by the date filter: {'birth': 116, 'visit': 36}
One row per mouse and day (same-day weights averaged, rows without weight dropped):
3049 rows = len(df)
```

```
3049
```

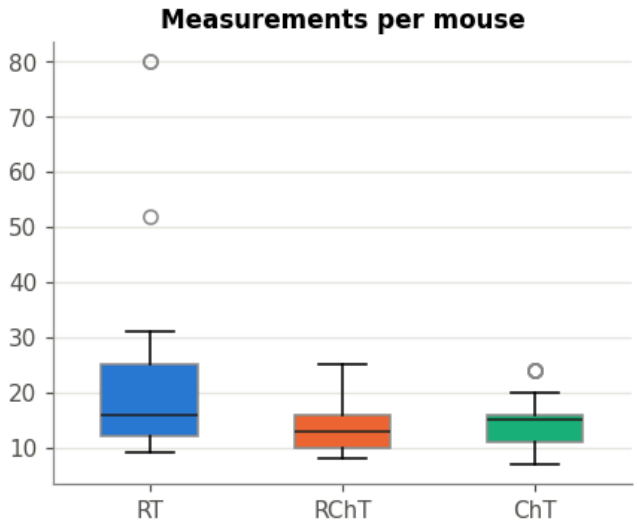
C2. Table 1: characteristics of the study cohort

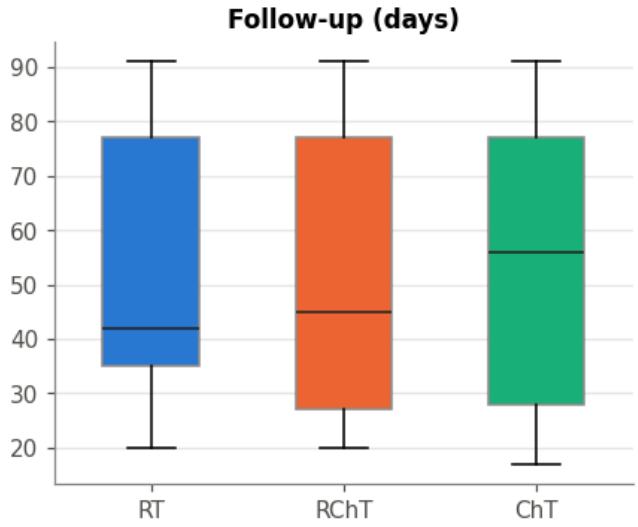
One column per therapy group, as in the results tables. Continuous values are median [interquartile range]; ages are at the tumour injection. *Baseline weight* is the first measurement of the series.

	Radiotherapy (RT)	Radio- and chemotherapy (RChT)	Chemotherapy (ChT)	Overall
Mice	73	60	53	186
Sex: male	44 (60%)	42 (70%)	36 (68%)	122 (66%)
Sex: female	29 (40%)	18 (30%)	17 (32%)	64 (34%)
Strain: PFP/RAG2	19 (26%)	20 (33%)	19 (36%)	58 (31%)
Strain: SCID	54 (74%)	40 (67%)	34 (64%)	128 (69%)
Cell line: OH-1 (small-cell lung cancer (SCLC))	38 (52%)	24 (40%)	19 (36%)	81 (44%)
Cell line: H69 (small-cell lung cancer (SCLC))	16 (22%)	16 (27%)	15 (28%)	47 (25%)
Cell line: LNCaP (prostate cancer (androgen-sensitive))	19 (26%)	20 (33%)	19 (36%)	58 (31%)
Age at tumour injection (weeks)	18.6 [17.4–20.9]	18.7 [16.7–22.3]	18.7 [16.7–20.9]	18.6 [16.7–21.5]
Baseline weight (g)	24.8 [22.6–27.1]	26.7 [23.8–29.4]	25.2 [22.6–27.1]	25.6 [22.8–27.8]
Weight at last measurement (g)	23.0 [20.3–25.5]	22.8 [19.8–25.6]	24.3 [20.7–26.7]	23.3 [20.3–25.9]
Measurements per mouse	16 [12–25], range 9–80	13 [10–16], range 8–25	15 [11–16], range 7–24	15 [11–16], range 7–80
Follow-up, first to last measurement (days)	42 [35–77]	45 [27–77]	56 [28–77]	42 [28–77]
RT: fractions per mouse	5 [5–5]	5 [5–5]	–	5 [5–5]
RT: total dose (Gy)	50 [50–50]	50 [50–50]	–	50 [50–50]
ChT: Cisplatin	0 (0%)	40 (67%)	34 (64%)	74 (40%)
ChT: Docetaxel	1 (1%)	20 (33%)	19 (36%)	40 (22%)
ChT: administrations per mouse	1 [1–1]	1 [1–3]	1 [1–3]	1 [1–3]
Days from injection to first treatment	20 [15–42]	34 [20–49]	35 [20–49]	28 [15–49]
Days from injection to death	35 [29–70]	43 [28–70]	51 [28–78]	43 [28–70]
Death: Experiment ended as planned	60 (82%)	42 (70%)	51 (96%)	153 (82%)
Death: Killed: poor general condition	5 (7%)	6 (10%)	0 (0%)	11 (6%)
Death: Killed: tumour ulceration	0 (0%)	0 (0%)	1 (2%)	1 (1%)
Death: Spontaneous death	3 (4%)	4 (7%)	1 (2%)	8 (4%)
Death: Unknown	5 (7%)	3 (5%)	0 (0%)	8 (4%)
Death: Unknown: disposed of by animal care	0 (0%)	5 (8%)	0 (0%)	5 (3%)

C3. Measurement process

How many weights each mouse contributes, over how many days, and how irregular the spacing is. The models see the date of each weight (days since the first visit, seasonal features), so the spacing matters.



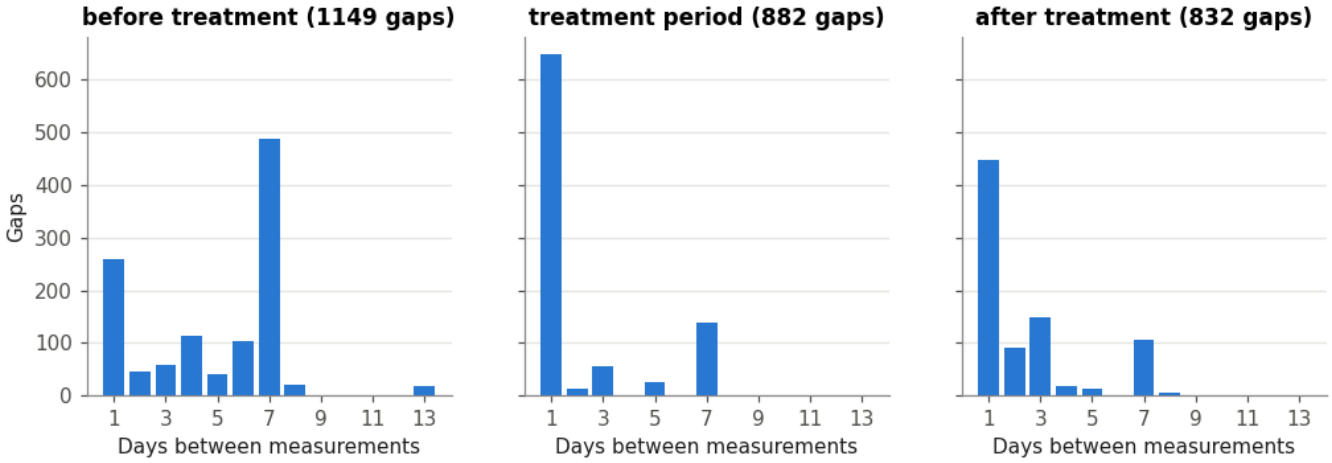


role	mice	measurements	per_mouse	follow_up_days
RT	73	1444	16 [12–25]	42 [35–77]
RChT	60	837	13 [10–16]	45 [27–77]
ChT	53	768	15 [11–16]	56 [28–77]

Mice with more than 40 measurements:

item_id	role	n	first	last	follow_up_days	cell_line
380	1	80	2016-05-09	2016-08-08	91	OH-1 RGB
392	1	52	2016-05-09	2016-07-11	63	OH-1 RGB
423	1	80	2016-05-09	2016-08-08	91	OH-1 RGB

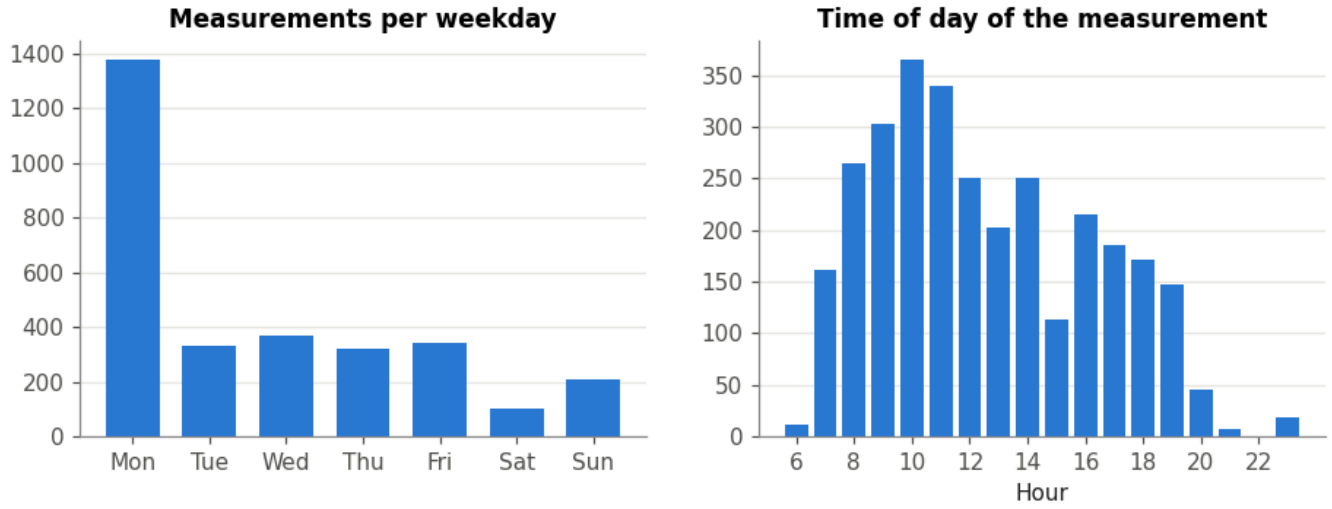
Gaps between consecutive measurements, overall and by treatment phase. The phase of a gap is that of its later measurement: *before treatment* (before the first treatment day), *treatment period* (first to last treatment day) and *after treatment*.



phase	gaps	median	mean	max
before treatment	1149	6.0	4.899043	13.0
treatment period	882	1.0	2.205215	7.0
after treatment	832	1.0	2.409856	8.0

gap	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	10.0	13.0
% of all gaps	47.4	5.2	9.2	4.6	2.8	3.6	25.6	0.9	0.0	0.6

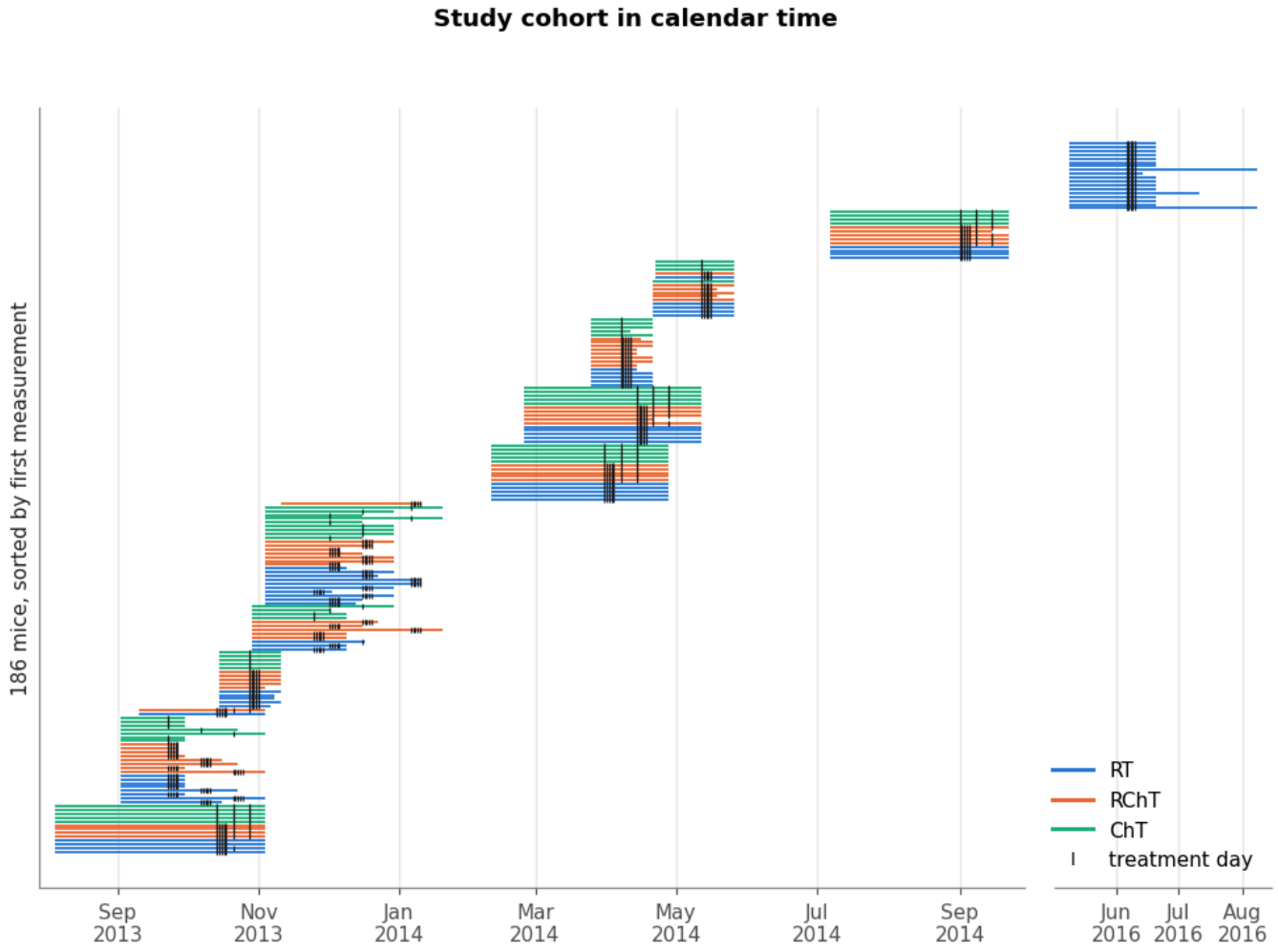
'47.4% / 25.6%'



Before treatment the mice are weighed about once a week (median gap 6 days, most often 7 days, typically on Mondays). From the first treatment on they are weighed almost daily (median gap 1 day). The sampling density therefore changes exactly where the weight starts to fall.

Calendar view of the cohort: one line per mouse from its first to its last measurement, small ticks on treatment days. Mice that overlap in calendar time

can serve as context for each other in the trail-context model. Long gaps without any mouse are cut out of the time axis.



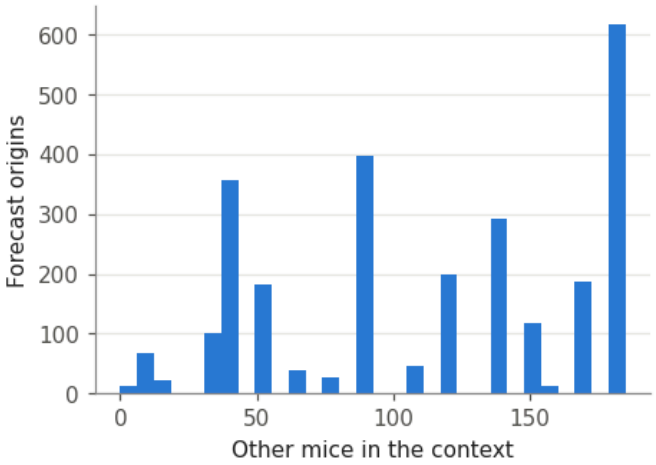
year of first measurement	therapy	ChT	RChT	RT	Total
	2013	30	31	31	92
	2014	23	29	24	76
	2016	0	0	18	18
	Total	53	60	73	186

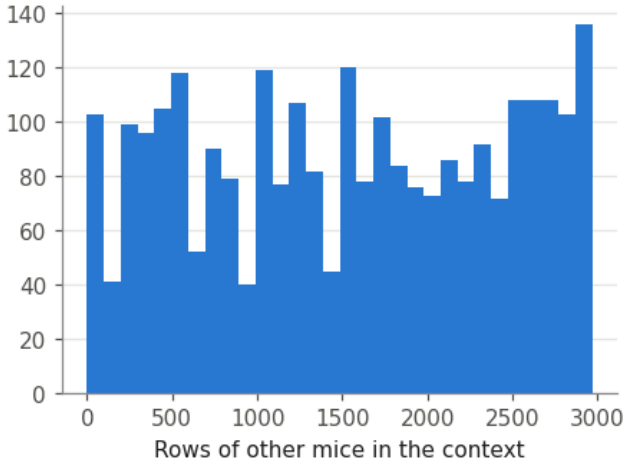
role	mice	measurements	per_mouse		cell_line
RT	18	671	31	[31–31]	OH-1 RGB

Most mice were measured between August 2013 and September 2014, in overlapping injection batches. The 18 mice of the May 2016 OH-1 batch are all RT mice and were weighed daily. With up to 80 measurements each they hold the longest series of the cohort, and they hardly overlap in time with any other mouse.

Size of the trail context. For every forecast origin of the rolling evaluation, the trail-context model sees the other cohort mice with more than `MIN_CONTEXT` measurements up to the date of the last own measurement (`context.pool_rows`). Below: how many mice and rows that pool holds.

```
186 mice -> 2677 forecast origins
```





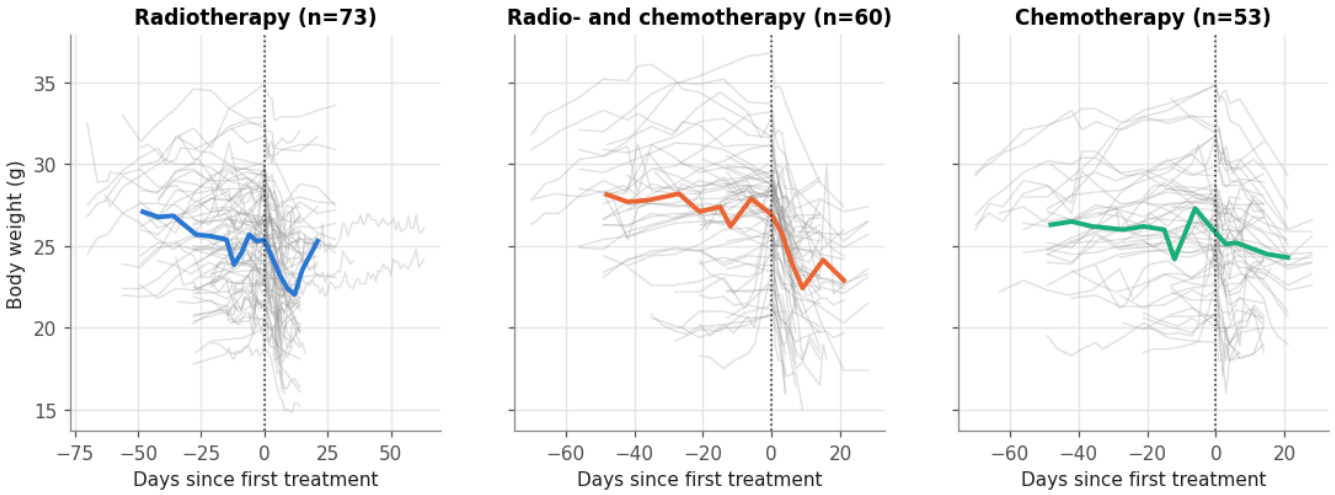
	count	mean	std	min	25%	50%	75%	max
pool_mice	2677.0	113.2	57.8	0.0	52.0	121.0	167.0	185.0
pool_rows	2677.0	1539.4	885.6	0.0	740.0	1552.0	2325.0	2968.0

```
Origins without any other mouse in the context: 13 of 2677
```

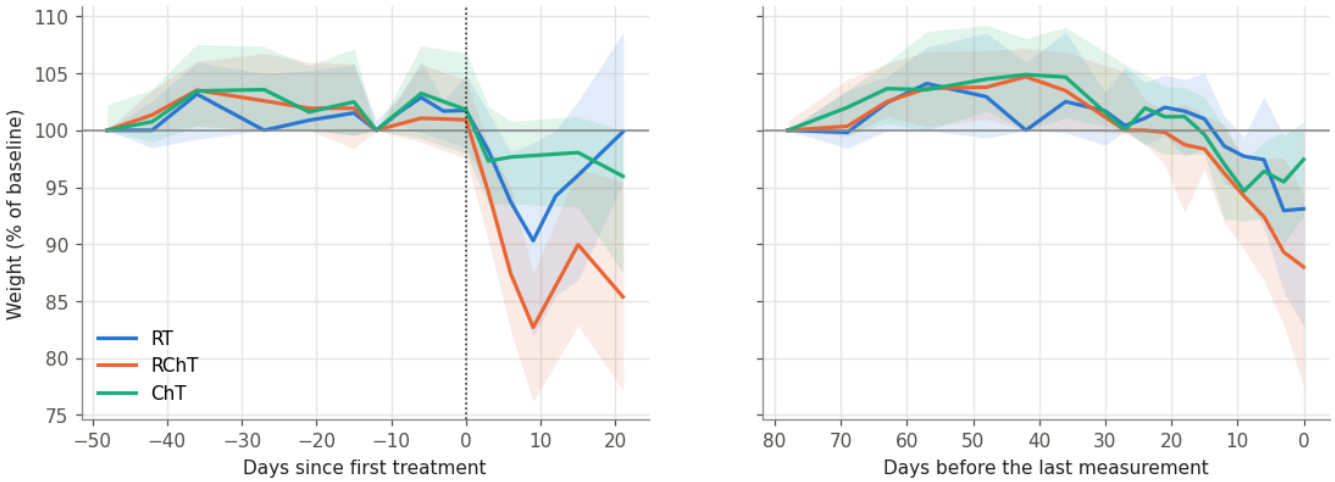
The pool only contains mice measured on or before the forecast date. It therefore grows with calendar time, from almost empty for the first batch to all other mice for the 2016 batch, which sees the complete earlier lives of the 2013–2014 mice.

C4. Weight trajectories

All series per therapy group, aligned at the first treatment day (day 0). Thin grey lines are single mice. The coloured line is the median over mice in 3-day bins centred on day 0, ±3, ±6, ... Each mouse counts once per bin (its mean), so the daily-weighed mice do not dominate; bins with fewer than 15 mice are not drawn.



Relative weight (% of the baseline weight) of the three groups on one axis, aligned at the first treatment (left) and at the last measurement (right). Median with interquartile range.

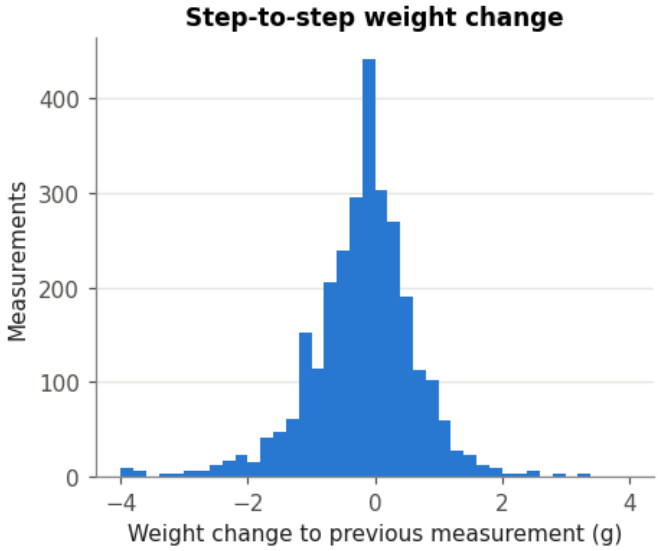


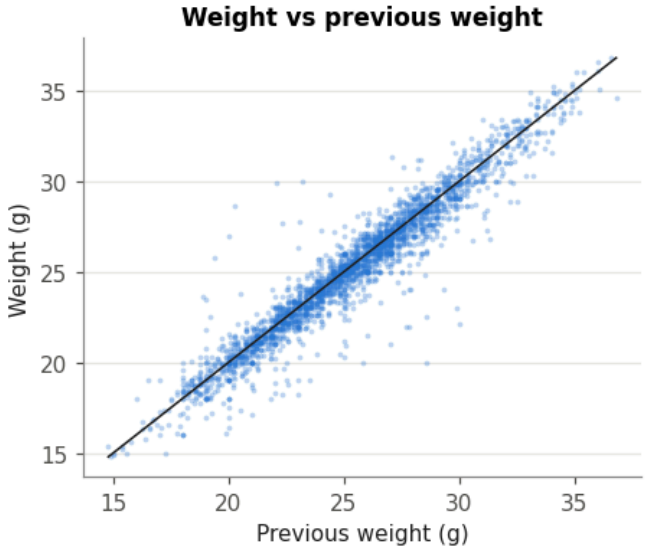
Before treatment the weights are stable or rise slightly. After the first treatment the weight falls in all groups: most strongly and for longest after radiochemotherapy, least after chemotherapy alone. Only about 20 mice per group are followed for more than three weeks after the first treatment, so the right end of the left panel rests on few mice. Aligned at the end (right panel), the weight falls over the last two weeks of a series, which matters for the `days_before_death` analysis of the results.

Weight change between consecutive measurements (the quantity a one-step forecast has to get right), by phase and by the length of the gap.

phase	changes	mean	sd	median	q05	q95
before treatment	1149	0.11	0.74	0.1	-1.00	1.20
treatment period	882	-0.40	1.14	-0.3	-1.99	0.90
after treatment	832	-0.26	1.13	-0.1	-2.20	1.14

gap (days)	changes	mean_abs	sd
1	1356	0.55	0.82
2–3	413	0.94	1.39
4–6	317	0.77	1.31
7	732	0.65	0.89
>7	45	0.72	0.83





Correlation weight vs previous weight: 0.962
Correlation weight change vs previous weight change: -0.122
SD of the mean weight between mice: 3.35 g, SD of the weight within a mouse: 1.66 g

Consecutive weights are highly correlated, so the last weight is already a strong forecast (see C7). Consecutive changes are slightly negatively correlated: a jump tends to be partly undone at the next visit, which is typical of measurement noise (food, faeces, time of day). Mice differ more from each other than a mouse varies over its life, so the level of a mouse is easy and its change is the hard part. The changes are largest, and mostly negative, during the treatment period.

Baseline weight, lowest weight and weight at the last measurement by sex and cell line.

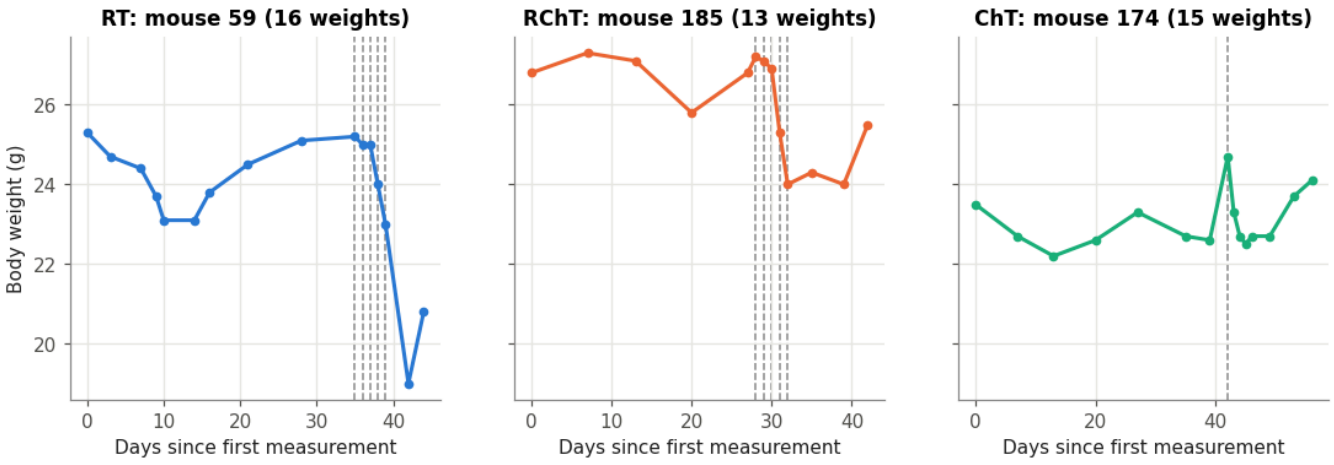
sex	mice	baseline_g	lowest_pct_of_baseline	last_weight_g
female	64	22.2 [20.6–23.6]	86.1 [80.6–90.2]	19.8 [18.2–21.4]
male	122	27.1 [25.6–29.0]	91.8 [83.0–96.7]	25.2 [23.1–26.8]

cell_line	mice	baseline_g	lowest_pct_of_baseline	last_weight_g
H69	47	25.7 [22.4–27.6]	87.5 [83.4–92.9]	23.7 [20.2–25.6]
LNCaP	58	26.4 [24.7–28.2]	94.0 [82.8–98.4]	24.6 [23.3–26.3]
OH-1	81	24.6 [22.2–27.4]	86.8 [79.9–94.2]	21.5 [19.6–25.4]

cell_line	sex	mice	baseline_g	lowest_pct_of_baseline	last_weight_g
H69	female	21	22.2 [20.6–23.8]	85.1 [82.1–89.6]	20.1 [18.4–21.9]
	male	26	27.6 [26.9–28.4]	88.0 [85.3–93.5]	25.5 [24.3–26.6]
LNCaP	male	58	26.4 [24.7–28.2]	94.0 [82.8–98.4]	24.6 [23.3–26.3]
OH-1	female	43	22.2 [20.6–23.5]	86.1 [80.3–90.4]	19.7 [18.2–21.1]
	male	38	27.5 [26.3–29.5]	89.5 [79.1–96.6]	25.9 [22.4–27.1]

Female mice are about 5 g lighter than male mice and lose relatively more weight. All LNCaP (prostate) mice are male.

Three example mice, the one with the median number of measurements in each group, with the treatment days marked.



C5. Treatments within the series

After preprocessing each treatment day is one row with `intervention = 1` (radio and chemo counted separately). The position of the first treatment tells how much own history a mouse has before the treatment starts.

role	mice	rows	rows_with_intervention	RT_days_per_mouse	ChT_days_per_mouse	measurements_before
RT	73	1444	362	5 [5–5]	0 [0–0]	
RChT	60	837	338	5 [5–5]	1 [1–3]	
ChT	53	768	91	0 [0–0]	1 [1–3]	

role	% of forecast origins: context has treatment	% of forecast origins: target is treatment day
RT	56.9	27.9
RChT	57.9	47.1
ChT	56.6	13.7

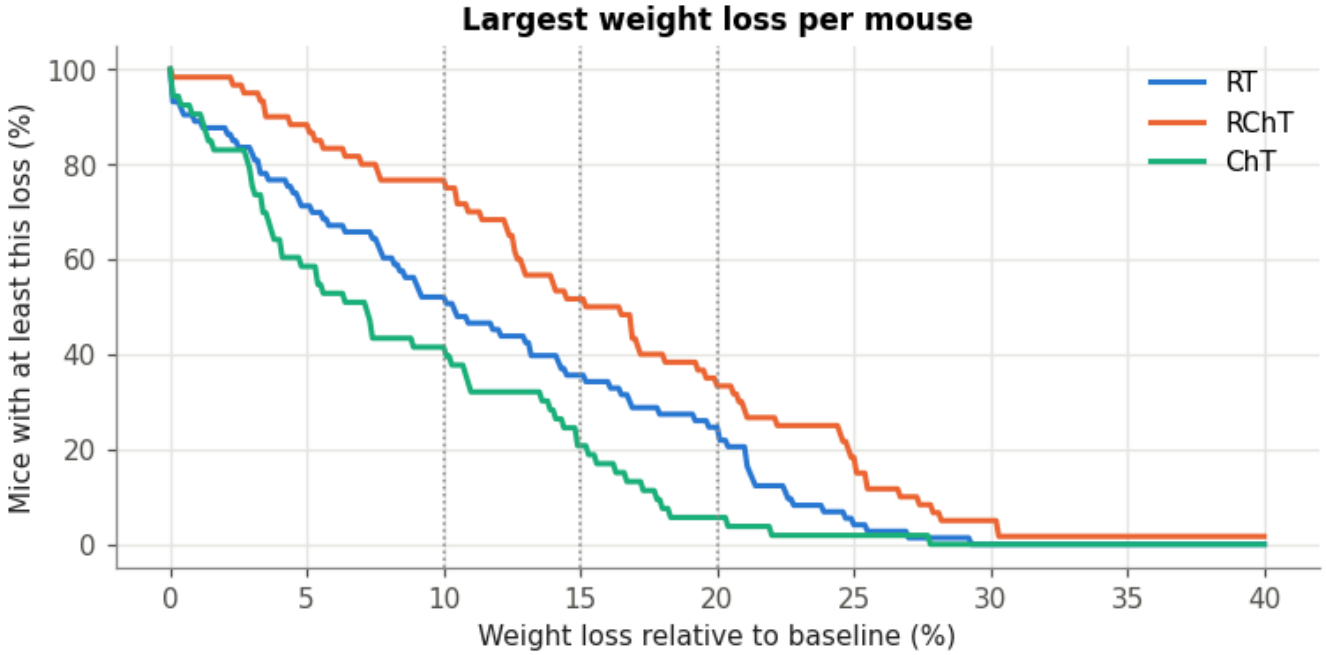
A quarter of all rows are treatment days. Each mouse has a median of 6–7 weights before its first treatment. Just over half of the forecast origins already have a treatment in their own context, and in the RChT group almost half of the forecast targets are treatment days themselves.

C6. Weight loss and endpoints

Lowest weight relative to the baseline, how many mice crossed common weight-loss limits, how this relates to the cause of death, and how close the last measurement is to death.

role	mice	lowest_weight_pct_of_baseline	loss_at_least_10pct	loss_at_least_15pct	loss_at_least_20pct
RT	73	89.7 [80.3–95.6]	38 (52%)	26 (36%)	18 (25%)
RChT	60	84.2 [77.2–89.7]	46 (77%)	31 (52%)	20 (33%)
ChT	53	92.8 [85.7–97.0]	22 (42%)	11 (21%)	3 (6%)

death_cause	Experiment ended as planned	Killed: poor general condition	Killed: tumour ulceration	Spontaneous death	Unknown	Unknown: disposed of by animal care	Total
loss_group							
< 10%	76	1	0	1	2	0	80
10–15%	29	1	0	2	4	2	38
15–20%	22	1	1	3	0	0	27
≥ 20%	26	8	0	2	2	3	41
Total	153	11	1	8	8	5	186



days from last measurement to death	0	1	2	3	4	5	6	7
mice	42	125	1	9	6	1	1	1

```
'1 [1-1]'
```

The last measurement is almost always one day before death, or on the day itself. The `days_before_death` axis in the results (`rolling.py`) counts from the last measurement, which is therefore a close stand-in for the day of death.

About a fifth of the cohort lost at least 20% of its baseline weight at some point, almost all of them in the irradiated groups (RT and RChT). Even so, most of these mice still ended the experiment as planned: weight loss alone rarely ended a series early.

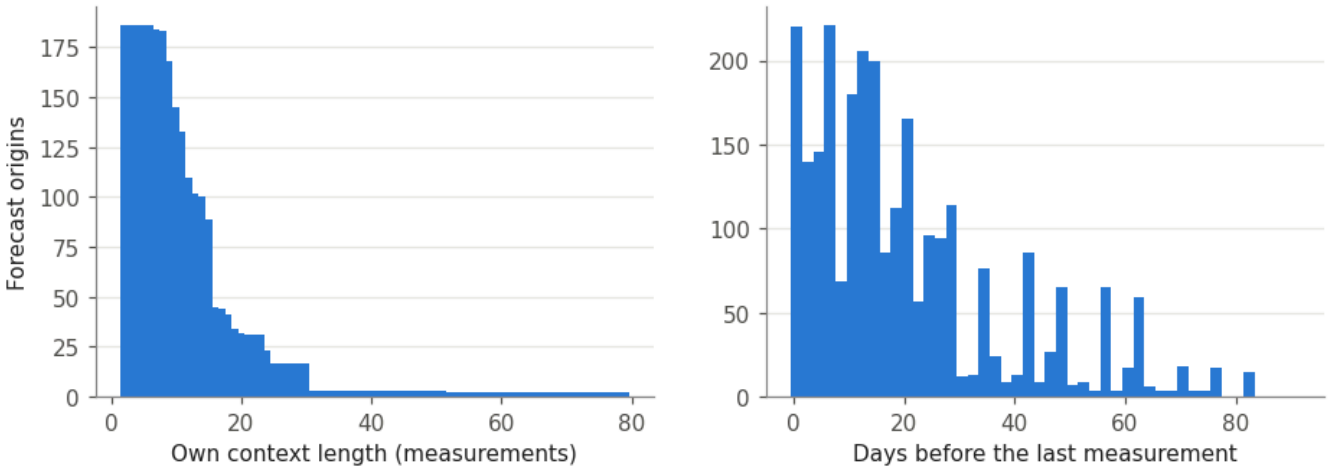
C7. Evaluation footprint

The rolling one-step-ahead evaluation predicts every measurement from the `MIN_CONTEXT + 1`-th on from all earlier weights of the same mouse. Number of forecast origins per group and how much own context they have.

role	mice	forecast_origins	context_length	gap_to_target_days
RT	73	1298	11 [6–20]	1 [1–3]
RChT	60	717	7 [4–11]	3 [1–7]
ChT	53	662	8 [5–12]	3 [1–7]
Overall	186	2677	9 [5–14]	1 [1–7]

The 18 RT mice of the 2016 batch account for 635 of the 1298 RT forecast origins (49%) and 24% of all origins.

```
Forecast origins = rows − 2 × mice = 3049 − 2 × 186 = 2677
Cached forecasts in /data/PFN/Mouse/rolling/tabpfn_trail3.parquet: 2677 rows, same origins as here: True
```



For scale: the error of the naive forecast "next weight = last weight" on exactly these origins, computed with the metric functions of the results notebook (`rolling_metrics`: mean over origins per group, *Overall* is the mean of the three groups). This is a reference point for reading the model errors, not one of the models of the paper.

```
186 mice -> 2677 forecast origins
```

model	role	n	MAE	MAPE
Last value (naive reference)	**Chemotherapy**	662	0.7008	0.0283
	Overall	2677	0.6953	0.0284
	Radio- and chemotherapy	717	0.8435	0.0344
	Radiotherapy	1298	0.5415	0.0226

The metrics weight every forecast origin equally within a group. In the RT group the 18 daily-weighed mice of 2016 contribute a large share of the origins, with short gaps and therefore small changes, which is one reason why the naive error is lowest for RT.

C8. Data quality in the cohort

What the preprocessing changed on the way from the raw visits to the series.

Mouse-days with more than one visit (weights averaged): 10

days to the nearest visit	0	1
treatment		
chemo	188	2
radio	660	1

A treatment takes over the weight of its nearest visit; that visit is removed, so i
ts weight moves to the treatment date.
Visits absorbed this way: 790

Mice excluded for implausible weights (raw visits):

	id	date	weight	score
1673	82	2013-10-23	17.0	5
1702	82	2013-10-24	17.0	5
1708	82	2013-10-25	15.0	15
1741	82	2013-10-28	0.0	0

	id	date	weight	score
4843	313	2014-05-15	22.0	5
4867	313	2014-05-16	22.0	5
4914	313	2014-05-19	21.5	5
4959	313	2014-05-26	222.2	5

Two loaders exist. The results use `michael.load_dataset` (the preprocessing behind the Eagle-eye baseline, MIE 2025). `preprocessing.load_dataset` keeps all visits and attaches treatments to the nearest visit instead of moving the weight. Same configuration, for comparison:

```
Animals: 188 -> 186 after filtering -> 186 sampled
```

	michael.load_dataset (used)	preprocessing.load_dataset
mice	186	186
rows	3049	3084
rows with intervention	791	790
first date	2013-08-05	2013-06-17

Part D: facts for the paper

The key numbers collected above, in one place.

	value
Tables in the database	25
Animals in the database	843
Visits in the database	22142
Trails (experiment series)	15
Approved projects	12
Mice in trail 3	376
Trail 3: mice killed at the planned end of the experiment	296 (79%)
Study cohort: mice	186
Study cohort: daily weight measurements (rows)	3049
Gap between measurements: 1 day / 7 days (% of gaps)	47.4% / 25.6%
Rows with a treatment (intervention = 1)	791 (26%)
Days from last measurement to death	1 [1–1]
RT forecast origins from the daily-weighed 2016 batch (18 mice)	635 (49%)
Forecast origins in the rolling evaluation	2677
Naive last-value forecast: MAE overall (g)	0.695
Study cohort: RT / RChT / ChT mice	73 / 60 / 53
Study cohort: male	122 (66%)
Study cohort: cell lines OH-1 / H69 / LNCaP	81 / 47 / 58
Study cohort: age at tumour injection (weeks)	18.6 [16.7–21.5]
Study cohort: baseline weight (g)	25.6 [22.8–27.8]
Study cohort: measurements per mouse	15 [11–16], range 7–80
Study cohort: follow-up (days)	42 [28–77]
Study cohort: days from injection to first treatment	28 [15–49]
Study cohort: days from injection to death	43 [28–70]
Study cohort: killed at the planned end	153 (82%)
Study cohort: weight loss ≥ 20 % of baseline	41 (22%)
Study cohort: calendar period	2013-08-05 to 2016-08-08